

Sai Malla

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Summary

AI/ML Engineer with 5+ years of experience building ML, NLP, deep learning, and Generative AI applications using Python, PyTorch, TensorFlow, Scikit-learn, and PySpark. Experienced with LangChain, LangGraph, MCP, RAG, FAISS, Pinecone, GPT-4o, Llama 3, and Mistral, including agent workflows, tool calling, prompt engineering, and LLM evaluation. Strong background in FastAPI, AWS, Docker, Kubernetes, MLflow, LangSmith, Airflow, and CI/CD, with experience deploying and monitoring production of AI/ML systems.

Experience

AI/ML Engineer

Apr 2024 – Present

Langchain | CA, USA

- Owned the design and development of production-grade LLM and AI/ML applications in Python using PyTorch, Scikit-learn, Pandas, and NumPy, driving end-to-end implementation across agent workflows, NLP, model evaluation, and developer-facing AI applications.
- Built RAG applications with LangChain, integrating document loaders, chunking, embeddings, retrievers, reranking, and vector stores such as FAISS and Pinecone for grounded responses.
- Integrated GPT-4o, Llama 3, and Mistral with LangChain, improving agent behavior through prompt engineering, structured outputs, query rewriting, context management, and model selection.
- Built stateful agent workflows with LangGraph, coordinating retrieval, reasoning, validation, and tool execution with state management, conditional routing, retries, and human-in-the-loop flows.
- Implemented MCP and LangChain tool calling to connect agents with APIs, databases, Python functions, search tools, and external services, including tool validation and execution error handling.
- Developed Python and FastAPI backend services for agent APIs and supporting services, using PostgreSQL, Redis, asynchronous processing, authentication, and REST APIs for scalable application workflows.
- Worked on agent and model evaluation, using LangSmith and MLflow to trace runs, compare prompts/models, analyze failures, and measure latency, relevance, faithfulness, tool-call success, and response quality.
- Built A/B testing and offline evaluation workflows to compare agent prompts, retrieval strategies, and model versions, using production feedback and evaluation datasets to guide improvements.
- Used Apache Airflow for scheduled data preparation, evaluation, and supporting ML workflows, while applying LoRA, QLoRA, PEFT, feature engineering, and hyperparameter tuning for selected ML/LLM tasks.
- Containerized and deployed services using Docker, Kubernetes, AWS, and CI/CD, with CloudWatch and LangSmith for monitoring; reduced agent latency by 25% and improved evaluation scores by 15–20%.

AI/ML Engineer

Aug 2020 – Jul 2023

Databricks | India

- Worked on a production recommendation platform using Python, Pandas, NumPy, PySpark, SQL, Databricks, and Delta Lake to process customer behavior, product information, and interaction data.
- Built and tuned recommendation, ranking, classification, and similarity models using Scikit-learn, Spark MLlib, XGBoost, and LightGBM, improving recommendation relevance by approximately 15% through iterative model optimization.
- Created reusable feature engineering and model-training pipelines with PySpark, Airflow, Databricks Jobs, and MLflow, reducing manual model-training effort by around 30%.
- Developed real-time inference services with FastAPI and REST APIs, connecting models with MongoDB and using Redis to cache frequently requested recommendations, reducing average API latency by approximately 35%.
- Worked on NLP and deep learning models using PyTorch, TensorFlow, Hugging Face Transformers, BERT, and sentence embeddings to improve product and user-interest understanding.
- Built semantic recommendation capabilities using embeddings and similarity search, allowing recommendations to consider user intent and contextual product information.
- Used SHAP and LIME to investigate model predictions and A/B testing to compare recommendation approaches using real user engagement signals.
- Implemented MLOps practices with MLflow, Docker, Kubernetes, Git, and CI/CD, supporting experiment tracking, model versioning, deployment, retraining, and production model management.
- Extended the platform with LLM and Generative AI capabilities using foundation models, prompt engineering, embeddings, FAISS, vector search, and RAG for semantic search and context-aware, grounded responses.
- Set up production AI/ML monitoring with Prometheus and Grafana, tracking model performance, data drift, inference latency, API health, and resource utilization, with Pytest for ML and API testing.

Technical Skills

Programming & Core Engineering: Python, C++ systems, Rust, SQL, Bash, Data Structures & Algorithms, Object-Oriented Programming, Multithreading, Async Programming (asyncio), Concurrency Control, Performance Optimization, System Design

LLM & Agents: LangChain, LangGraph, MCP, tool/function calling, agentic workflows, RAG, prompt engineering, structured outputs, query rewriting, context management, human-in-the-loop, guardrails, GPT-4o, Llama 3, Mistral

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Spark MLlib, XGBoost, LightGBM, Hugging Face Transformers, BERT, sentence embeddings, LoRA, QLoRA, PEFT, feature engineering, hyperparameter tuning, SHAP, LIME, model serving

Retrieval & Vector: FAISS, Pinecone, embeddings, chunking, retrievers, reranking, semantic search, similarity search, hybrid search, metadata filtering, vector databases

Evaluation: LangSmith, MLflow, LLM evaluation, offline evaluation datasets, A/B testing, faithfulness, relevance, tool-call success, regression testing, latency, cost and failure analysis

Data & Backend: Python, SQL, REST APIs, FastAPI, Pydantic, SQLAlchemy, Git, Pytest, PostgreSQL, MongoDB, Redis, Pandas, NumPy, PySpark, Databricks, Delta Lake, Apache Airflow, asynchronous processing, authentication, microservices

MLOps & Cloud: Docker, Kubernetes, AWS, SageMaker, S3, Lambda, ECR, EKS, CI/CD, MLflow, model versioning, retraining, Terraform, GitHub Actions, Prometheus, Grafana, CloudWatch, data drift monitoring

Education

Master of Science in Management Information Systems

University of Memphis